

S002 - Throughput and Resilience Programme: Implementation Plan

Implementation Plan | Engineering

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Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-07-28
Related Scenario ID	SC-003, SC-011	Document Type	Implementation Plan

1. Programme Overview

Field	Value
Programme name	S002 Throughput and Resilience
Programme code	PRJ-S002-TR
Seller	S002 - BluePeak Electronics
Trigger	SC-011 (June 2026) and SC-003 (July 2026)
Start	2026-07-28
Target completion	2026-11-20
Engineering owner	Vikram Singh
Operations owner	Neha Gupta
Seller counterpart	Sameer Khanna

2. Rationale

Two incidents within six weeks exposed the same underlying weakness from different directions.

In June 2026 the order event consumer entered a rebalance loop after a partition expansion, because the maximum poll interval was shorter than the time needed to process a batch. 6,700 events were stranded and the settlement feed was stale for nine hours.

In July 2026 an unbounded client retry policy during a flash sale, combined with a missing circuit breaker on the inventory lookup, exhausted the order service connection pool. p99 latency reached 12.4 seconds and 3,900 order placements failed.

Neither incident was caused by insufficient hardware. In both cases the system lacked a mechanism to degrade in a controlled way when a component fell behind. This programme adds those mechanisms and validates them under load before the festive peak.

3. Workstreams

ID	Workstream	Owner	Target	Status
WS-A	Load shedding and backpressure	Vikram Singh	2026-08-28	Complete
WS-B	Consumer stability and lag control	Vikram Singh	2026-09-18	In progress

ID	Workstream	Owner	Target	Status
WS-C	Capacity model and flash-sale readiness	Neha Gupta	2026-10-09	In progress
WS-D	Client integration hardening at BluePeak	Sameer Khanna	2026-10-23	Not started
WS-E	Peak readiness validation	Neha Gupta	2026-11-20	Not started

3.1 WS-A Load Shedding and Backpressure

Delivered on 26 August 2026, two days early.

- Circuit breaker on the inventory lookup path, opening at a 25 percent error rate over a 10-second window, with a 30-second half-open probe.
- Connection pool resized from 120 to 300 with an explicit acquisition timeout of 750 ms, so a saturated pool fails fast rather than queueing.
- Overload responses changed from HTTP 503 to HTTP 429 with a Retry-After header, making the correct client behaviour explicit.
- Request-level timeout of 2.5 seconds on the order placement path.

Validation: a synthetic retry storm at four times observed July peak was run on 24 August 2026. The service shed load at 429 and recovered within 40 seconds of the storm ending, with no pool exhaustion.

3.2 WS-B Consumer Stability and Lag Control

In progress, on track for 18 September 2026.

- Maximum poll interval increased to 300 seconds and batch size reduced to 200 records, so a worst-case batch completes well inside the interval.
- Processing moved off the poll thread, so a slow downstream call can no longer delay a heartbeat.
- Consumer lag published as a first-class metric per partition, with alerting above 15 minutes as required by ORD-4.
- A rebalance-loop detector that alerts when a consumer group performs more than three rebalances in five minutes.

Remaining: partition-level lag dashboards and the operator runbook entry for a stuck consumer group.

3.3 WS-C Capacity Model and Flash-Sale Readiness

A capacity model expressing order throughput as a function of concurrent sessions, cart size and inventory lookup fan-out. The model output is a readiness statement: for a declared flash-sale profile, either the current capacity is sufficient, or a specific scaling action is required.

The 10-working-day capacity notice is now a contractual expectation on this account, so the model has a defined input point rather than depending on informal warning.

3.4 WS-D Client Integration Hardening

BluePeak owns this workstream. Scope agreed on 21 July 2026:

- Replace the unbounded retry policy with a maximum of three attempts using exponential backoff with jitter.
- Honour Retry-After on 429 responses.
- Add a client-side circuit breaker so a platform degradation does not become a self-inflicted retry storm.
- Distinguish retryable from non-retryable failures in the client.

3.5 WS-E Peak Readiness Validation

A joint validation exercise in November 2026 covering a simulated flash sale, a forced consumer rebalance and a partial inventory dependency failure, with operations executing the Service Recovery Procedure against each.

4. Dependencies

Dependency	Owner	Needed by	Status
Partition expansion completed	Engineering	WS-B	Done, June 2026
Flash-sale calendar for Q4	BluePeak	WS-C	Received 2026-08-14
Client retry changes in BluePeak release train	BluePeak	WS-D	Scheduled October
Load test environment at 4x peak	Operations	WS-E	Provisioned

5. Risks

ID	Risk	Impact	Mitigation
R-1	WS-D slips past the festive freeze	High	Platform-side 429 handling limits blast radius even without client changes
R-2	Capacity model under-predicts a novel sale format	Medium	Readiness statement requires a named owner to accept residual risk
R-3	Lag alerting produces noise during normal batch peaks	Low	Threshold tuned against four weeks of observed lag
R-4	Change freeze blocks WS-B completion	Medium	WS-B targeted for 18 September, before the 10 October freeze

6. Success Criteria

1. A retry storm at four times July peak does not exhaust the connection pool.
2. Consumer lag never exceeds 15 minutes without an alert being raised.
3. A forced rebalance recovers without operator intervention.
4. A declared flash sale produces a written readiness statement at least five working days before the event.
5. No settlement feed staleness exceeding one hour during the festive peak.

7. Reporting

Fortnightly status to Operations and Business. Any change to the November peak readiness date is escalated immediately, because the festive freeze removes the option of late delivery.